

Phase 2 Results: Topology, Identity Mix, and Scoring Mode in a Climate-Firestorm Simulation

TUM M.Sc. Data and Society — thesis results chapter (standalone)

Harvest stamp: 19 September 2026

This chapter reports the Phase 2 campaign of the climate-firestorm simulation. The campaign crossed eight network topologies with homogeneous (H) and heterogeneous (He) identity mixes and with two auditor instruments, then compared a reconstructed Debnath hashtag graph against a custom-graph (D-net) simulation. Discrete MPR (dual-run headline) and continuous MPR (continuous-only headline) are different instruments; this chapter never averages them. The harvest is a real `gpt-4o-mini` run with $N = 1$, eight hops, and `thesisGrade=false`. Hatched dead cells remain in the grid. The Debnath comparison is structural: 814k tweet identifiers were not hydrated, and Debnath’s skip-gram window of eight is not this protocol’s hop budget.

Contents

List of Figures	2
List of Tables	3
1 Purpose of this chapter	5
2 Methods snapshot	5
2.1 Grid that was harvested	6
2.2 Shared protocol	6
2.3 Topologies and identity	7
2.4 Pfeffer knobs: swept, held, measured	7
2.5 Dead cells and k^*	7
3 Two scoring instruments	8
4 All eight topologies	8
5 Homogeneous versus heterogeneous identity	9
5.1 Homogeneous personas	10

5.2	Heterogeneous mixes	11
6	Four comparison designs	12
6.1	Design A: linear-chain analogue	14
6.2	Design B: networked topology and k^*	15
6.3	Design C: seed valence	15
6.4	Design D: closed-group wiring	16
7	Pooled-mean check	16
8	D-net, Pfeffer observables, and Debnath	17
8.1	Structural comparison	17
8.2	Simulated D-net auditor scores	18
8.3	Pfeffer 2014: seven factors, then a remapping	18
9	Limitations	19
10	Chapter conclusion	20
	References	21

List of Figures

1	Methods schematic for Phase 2	6
2	Mean dual gap (discrete headline minus sidecar continuous) by topology and arm. The gap is a scoring diagnostic, not a ground-truth MPR. $N = 1$, <code>gpt-4o-mini</code> , eight hops. Source: <code>results_phase2/figures/fig10_dual_gap_diagnostic.png</code> . If <code>analysis_full</code> later replaces this panel, keep this caption number.	9
3	Live-cell mean <code>meanMI</code> by topology, scoring mode, and arm. Left: homogeneous. Right: heterogeneous. Discrete (T2d) and continuous (T2c) bars are not averaged. Dashed line in later figures marks the propaganda threshold $MI = 3$; this panel uses the raw 0–5 scale. $N = 1$. Source: <code>fig01_topology_mpr_discrete_vs_continuous.png</code>	10
4	Share of live cells with irreversible network k^* (mean $MI > 3$ with no later recovery), by topology and scoring mode. Dual k^* uses the discrete series; continuous k^* uses the continuous series. The two rates are not added. Right-censored at eight ticks. $N = 1$. Source: <code>fig09_kstar_rates.png</code>	10
5	Hatched dead-cell rate by topology, arm, and scoring mode. Dead means <code>nScored</code> ≤ 1 after real LLM calls. Hatched cells are kept and are not scored as mix immunity. $N = 1$. Source: <code>fig08_dead_cell_rates.png</code>	11
6	Collapsed live-cell <code>meanMI</code> for homogeneous (H) versus heterogeneous (He) arms, separately for continuous (T2c) and dual-discrete (T2d) headlines. The two panels are not averaged. $N = 1$. Source: <code>fig02_h_vs_he_mpr.png</code>	11

7	Homogeneous personas: live-cell mean meanMI collapsed across eight topologies, discrete versus continuous. $N = 1$, 48 cells per persona per mode before hatching. Source: fig03_persona_mpr_bars.png	12
8	Persona families on homogeneous cells: conspiracy ($n = 4$), climate action ($n = 3$), science/environment ($n = 5$). Discrete and continuous remain separate. $N = 1$. Source: fig12_persona_family.png	12
9	Homogeneous persona $\times$ article heatmaps of live-cell meanMI . Left: continuous. Right: dual-discrete. The two panels are not averaged. $N = 1$. Source: fig05_persona_article_heatmaps.png	13
10	Homogeneous persona $\times$ topology heatmaps of live-cell meanMI . Left: continuous. Right: dual-discrete. $N = 1$. Source: fig06_persona_topology_heatmaps.png	23
11	Heterogeneous mixes: live-cell mean meanMI collapsed across eight topologies, discrete versus continuous. $N = 1$, 48 cells per mix per mode before hatching. Source: fig04_mix_mpr_bars.png	23
12	Heterogeneous mix $\times$ topology heatmaps of live-cell meanMI . Left: continuous. Right: dual-discrete. $N = 1$. Source: fig07_mix_topology_heatmaps.png	23
13	Empirical hashtag-graph metrics versus mean simulated D-net metrics (16 cascades). This is a structural comparison, not an MPR or Twitter-MI claim. Source: fig11_dnet_structural.png	24

List of Tables

1	Phase 2 harvest grid	6
2	Live-cell headline MPR by arm and instrument. Dead cells excluded. $N = 1$. Do not pool the two columns or the two rows.	8
3	Homogeneous (H) topology headlines. Each topology has 72 cells per mode (12 personas $\times$ 6 articles). Live means exclude hatched dead cells.	13
4	Heterogeneous (He) topology headlines. Each topology has 36 cells per mode (6 mixes $\times$ 6 articles).	13
5	Homogeneous personas collapsed across topologies. Family labels are analysis groupings, not Debnath HDBSCAN clusters.	14
6	Heterogeneous mixes collapsed across topologies.	14
7	Article (valence) collapse. Articles are a factor, not statistical replicates. $N = 1$	15
8	D-net structural metrics. Match = within a 30% band of the empirical value. KS/JS: $n_{\text{real}} = 1$, underpowered.	18
9	D-net auditor headlines on the hashtag co-occurrence graph. Sidecar continuous on dual runs is not the T2c headline.	18
10	Pfeffer observables on the Debnath hashtag fallback versus simulated D-net. Empirical valence is a toxicity/hashtag proxy, not MPR. Simulated mean MI is an auditor score, not Twitter MI.	19

1 Purpose of this chapter

This chapter reports the Phase 2 results of the climate-firestorm simulation. The purpose is to describe, under documented limits, how eight network topologies, homogeneous versus heterogeneous identity mixes, and two auditor instruments move misinformation scores on six geoengineering articles. The chapter is a thesis-style results account. It is not a reprint of the CIKM linear-chain study on commercial and crime news (Maurya et al., 2025), and it is not a Twitter digital twin of Debnath and colleagues’ 814,924-tweet observational map (Debnath et al., 2023b).¹

The theoretical spine remains online firestorms as defined by Pfeffer, Zorbach, and Carley (2014): a sudden discharge of large quantities of negative word-of-mouth in social media, often with high affective charge. Their Outlook section lists *seven* interrelated factors, not six. Section 8 restates those seven factors and records the thesis six-knob list as a computational remapping. Cross-media dynamics has no engine knob and is held.

Phase 2 asked a narrower empirical question than a full Pfeffer confirmation. Holding surprise as a drip seed and holding the clock at eight ticks, do topology, identity mix, and auditor mode change live-cell mean MI on climate seeds? The harvest answers that question only descriptively. `summary.json` sets `thesisGrade` to false. $N = 1$ with graph seed 42 forbids error bars and significance tests.

The remainder of the chapter follows a single path. Section 2 snapshots the protocol. Section 3 separates discrete from continuous MPR. Section 4 reports all eight topologies. Section 5 reports H versus He without pooling. Section 6 applies four comparison designs that match the previous paper’s coverage (chain analogue, networked k^* , identity composition, closed-group wiring) without reprinting crime-news numbers. Section 7 shows why pooled headlines mislead. Section 8 compares D-net structure and Pfeffer observables with the Debnath hashtag fallback. Section 9 states limitations. Section 10 closes.

Figures in this draft are generated from `thesisExperiment/results_phase2/figures/` (copies also under `analysis_phase2/figures/`). If a later `analysis_full` pass produces replacement panels, those files should be swapped in `figures/` without changing the caption numbers.

2 Methods snapshot

The Phase 2 campaign reused the existing CIKM engine entry point (`node index.js -config`) with thesis-layer configs under `thesisExperiment/configs/phase2/`. Each node is a persona-conditioned `gpt-4o-mini` agent. Allowed actions are forward, reinterpret, and drop. Propagation runs first; the auditor scores rewrites in a second pass (Maurya et al., 2025). The auditor is also `gpt-4o-mini`, so the scores are LLM-as-judge values in the sense of Zheng et al. (2023), not human ratings.

Figure 1 sketches the two-phase pipeline. A ground-truth article enters at the drip seed. Persona-conditioned nodes rewrite or forward the text along the graph. The auditor returns an

¹The `Paper/` folder referenced in the related-work notes is not present in this workspace. Coverage follows that note plus the CIKM engine in this repository: heatmaps, homogeneous versus heterogeneous identity, hop-limited trajectories, and a structural comparison, rewritten for Phase 2 rather than copied.

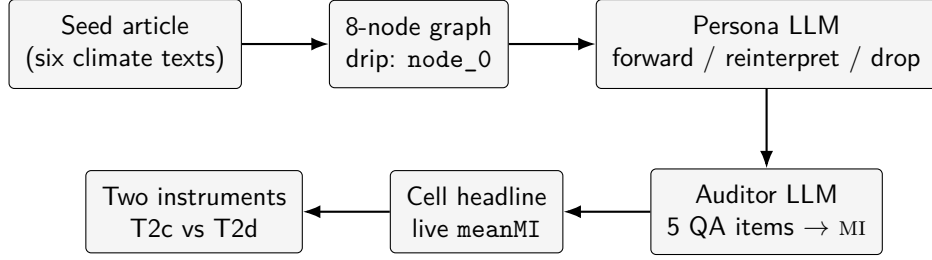


Figure 1: Methods schematic for Phase 2. A ground-truth climate article is rewritten by persona-conditioned `gpt-4o-mini` nodes on an eight-node graph, then scored by a five-item auditor. Discrete and continuous campaigns are separate configs. Network k^* is the first irreversible crossing of mean $MI > 3$, censored at eight ticks. This drawing is a protocol snapshot, not an empirical result.

Table 1: Phase 2 harvest grid. Source: `results_phase2/summary.json` (stamp 2026-09-19T05:27:59.674Z). Dead cells are hatched after real LLM calls (`nScored ≤ 1, llmCalls > 0`) and are excluded from live means.

Slice	Configs	Cells	Hatched dead	Status
T2c_H (cont., homo.)	96/96	576/576	94	complete
T2d_H (dual-disc., homo.)	96/96	576/576	98	complete
T2c_He (cont., hetero.)	48/48	288/288	50	complete
T2d_He (dual-disc., hetero.)	48/48	288/288	48	complete
D-net custom graph	4 configs	8 article-rows	0	complete
Thesis grid (T2 only)	288	1,728	290	<code>thesisGrade</code> false

An earlier parse note counted 292 hatched rows; the official harvest tables use 290. This chapter uses 290.

event MI on five yes/no items. Discrete MI equals the count of missing plus incorrect answers and therefore sits in $\{0, 1, 2, 3, 4, 5\}$. Continuous MI is a float on approximately the same 0–5 range. Node MPR is the mean of scored event MI for that node and article. The cell headline used below is parser `meanMI` (mean scored-event MI on live cells with `nScored > 1`). `meanNodeMPR` is the mean of node means of the same headline. The two scalars sit beside each other; they are not mixed into one number.

2.1 Grid that was harvested

Phase 2 isolated its outputs from Phase 1. Runs live under `thesisExperiment/runs_phase2/`; tables live under `thesisExperiment/results_phase2/`. The discrete 12×12 Phase 1 campaign in `runs/` and `results/tables/` was not overwritten and is not spliced into the headlines below.

Table 1 records the harvested grid. T2c denotes continuous-only scoring. T2d denotes dual scoring, whose *headline* is discrete. Homogeneous (H) cells cycle one of twelve personas on eight nodes. Heterogeneous (He) cells cycle one of six mixes of eight persona identifiers. Six core articles run sequentially in each config: SCoPEX 2017, chemtrails–Gates 2018–2021, SAI geoengineering, the Paris Agreement, climate consensus, and polar bears. The expected thesis grid is 1,728 cells. The harvest reports 0 missing cells.

2.2 Shared protocol

Every T2 cell used eight nodes, eight ticks, eight hops, inbox cap 4, trust threshold 0.15, action weights 0.35/0.50/0.15 (forward/reinterpret/drop), relation evolution on (`trustDelta`= 0.05),

activity pattern `always`, and `graphRandomSeed`: 42. The seed is always `node_0` (drip). The model is `gpt-4o-mini` for agent and auditor. Replicates: $N = 1$.

Eight hops is a logged cost cut versus the CIKM chain length of 30 (Maurya et al., 2025). It is not Debnath’s hop protocol. Debnath and colleagues define a skip-gram *context window of eight words* inside a tweet when they estimate p_{together} with “chemtrails” (Debnath et al., 2023b). That window is a word-embedding hyperparameter. It is not this campaign’s tick budget. The public R codes use an $n = 10$ n-gram implementation detail (Debnath, 2023); the paper text says eight.

2.3 Topologies and identity

The eight generators are `linear_chain`, `ring`, `random_er` (Erdős and Rényi, 1960), `small_world` (Watts and Strogatz, 1998), `scale_free` (Barabási and Albert, 1999), `echo_chamber`, `polarized`, and `hierarchical`. Linear chain is the CIKM object (Maurya et al., 2025). It is included so the climate corpus can be read against that topology, not because a chain is a firestorm clustering contrast. Scale-free uses Barabási–Albert $m = 2$. ER uses $p = 0.42$ with `minSeedOutDegree=2` so the drip seed cannot have out-degree zero.

Twelve homogeneous personas fall into three families: conspiracy (four), climate action (three), and science/environment (five). Belief profiles are theory-faithful reductions of Debnath discourse types, not HDBSCAN centroids on 814k tweets (Debnath et al., 2023b). Six heterogeneous mixes vary conspiracy share from four of eight (`mix_00`) to zero of eight (`mix_02`). `mix_02` is the expert-inclusive mix: climate-action, science, and environment identifiers only.

2.4 Pfeffer knobs: swept, held, measured

Pfeffer, Zorbach, and Carley (2014) do not name valence, surprise, identity, clustering, echo, and temporal acceleration as a canonical six-factor theory. The campaign treats those six names as engine knobs. Valence is varied by the six articles and by persona tone. Identity alignment is swept (H versus He). Network clustering is swept (eight topologies). Information echo is measured (homophily on the last-tick graph) and additionally wired in the echo-chamber and polarised generators. Surprise is held as drip seeding. Temporal acceleration is held at eight ticks. Cross-media dynamics, the sixth Outlook factor in 2014, has no broadcast agent and is held.

2.5 Dead cells and k^*

A cell is hatched dead when `nScored` ≤ 1 after real LLM calls. Dead cells are kept. They are not imputed as `MI=0` and they are not read as “mix immunises”. Live means drop those cells and report `n_live`.

Network k^* is the first tick at which network-mean `MI` > 3 (propaganda) and mean `MI` stays > 3 on every later tick with data. If the series later returns to ≤ 3 , k^* is undefined (none), not zero. The crossing is right-censored at eight ticks, so irreversibility is weak. Dual k^* uses the discrete series. Continuous k^* uses the continuous series. The two rates are not added.

Independent-cascade models treat adoption as a binary infection (Kempe, Kleinberg, and Tardos, 2003). This engine carries mutating text plus an auditor, so k^* clocks semantic degradation

Table 2: Live-cell headline MPR by arm and instrument. Dead cells excluded. $N = 1$. Do not pool the two columns or the two rows.

Arm	Cont. MPR (T2c)	n live cont.	Dual-disc. MPR (T2d)	n live dual
H (homogeneous)	0.8608	482	1.6814	478
He (heterogeneous)	1.0192	238	2.0800	240
He – H (same mode only)	0.1584	—	0.3986	—
Mean dual gap (sidecar)	—	—	0.8804 (H) / 1.1145 (He)	—

rather than a binary infected bit. The clock is still a simulation artefact on eight ticks, not an empirical firestorm duration.

3 Two scoring instruments

Phase 2 ran two auditor modes as separate campaigns. Continuous configs (prefix T2c) set `miScoringMode` to continuous. The headline MI/MPR is the continuous auditor float. Dual configs (prefix T2d) call both a discrete auditor and a continuous sidecar on the same events. The dual *headline* is the discrete 0–5 score. The sidecar continuous value is a scoring diagnostic. Dual gap (discrete minus sidecar continuous) is not a third ground-truth MPR.

These two headlines are not the same quantity. Dual-discrete H (1.6814) is not “the same MPR as continuous H (0.8608) but higher”. The auditors are different instruments. Dual-discrete sits higher than continuous on every topology in this harvest. That pattern is a scoring-mode effect. It is not evidence that the discrete campaign found more misinformation in a shared unit.

Table 2 reports live-cell means of `meanMI`. Dead cells are excluded. H and He remain separate. The harvest cross-check on H matches the recomputed live means to four decimals.

The same cells as `meanNodeMPR` (not interchangeable with `meanMI`) are H continuous 0.8595, H dual 1.6839, He continuous 1.0140, He dual 2.0737. This chapter leads with `meanMI` and treats `meanNodeMPR` as a sidecar of the same instrument.

Figure 2 plots mean dual gap by topology. The gap is large on every generator. Using it as an MPR would smuggle the discrete–continuous disagreement into a third league table. The figure is a diagnostic only.

4 All eight topologies

Topology is the clustering knob in the remapping of Pfeffer, Zorbach, and Carley (2014). Phase 2 swept eight generators rather than the Phase 1 pair (scale-free versus ER). Figure 3 shows live-cell mean `meanMI` for each generator, with discrete and continuous bars side by side and H and He in separate panels. Rank order is not the same across scoring modes. A single “best topology” from a pooled number is not available from this harvest.

Tables 3 and 4 list the same means with live counts, k^* rates, and dead rates. Linear chain and ring are short, low-degree paths. Hierarchical is a different degree sequence. With $N = 1$ the gaps are descriptive. They are not a significance claim about ER versus scale-free.

On homogeneous continuous cells, scale-free is highest (0.9205) and small-world is lowest (0.7215). On homogeneous dual-discrete cells, echo chamber is highest (1.7984) and small-world is lowest (1.5977). On heterogeneous continuous cells, hierarchical is highest (1.5242) and ER

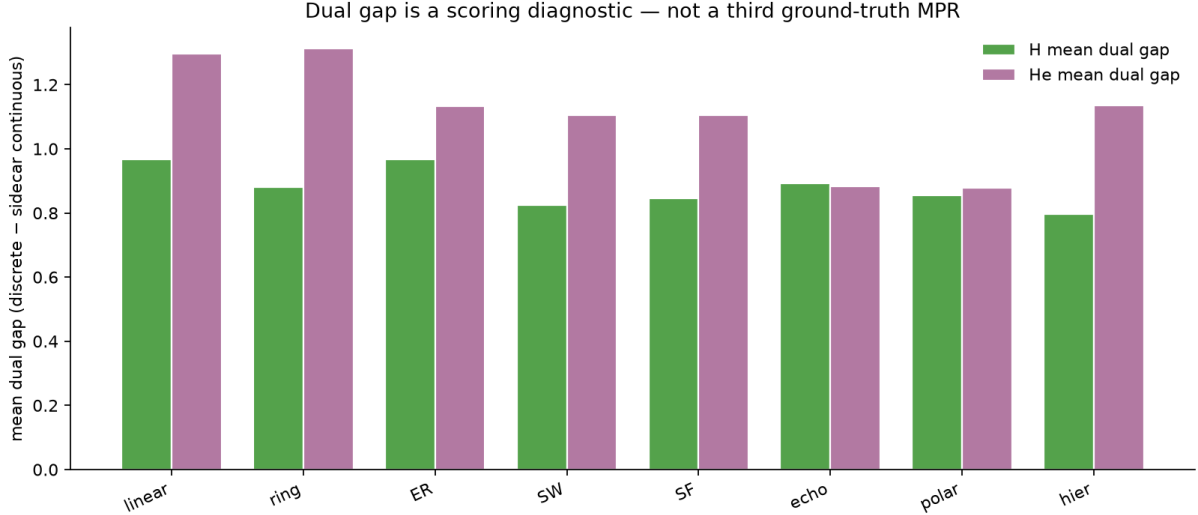


Figure 2: Mean dual gap (discrete headline minus sidecar continuous) by topology and arm. The gap is a scoring diagnostic, not a ground-truth MPR. $N = 1$, gpt-4o-mini, eight hops. Source: results_phase2/figures/fig10_dual_gap_diagnostic.png. If analysis_full later replaces this panel, keep this caption number.

is lowest (0.6593). On heterogeneous dual-discrete cells, hierarchical is highest (2.5934) and polarised is lowest (1.8830).

Hierarchical and polarised He cells also show high dead rates (about 28–31% continuous). A live-only mean on a half-dead topology inflates uncertainty. Those rows describe the surviving cascades, not a clean generator effect.

Figure 4 reports k^* occurrence on live cells. He dual-discrete k^* rates sit well above He continuous rates (32.1% versus 7.6% collapsed). The discrete instrument crosses and stays above 3 more often in this harvest. That is still a scoring-mode fact, not a second firestorm clock on a shared unit.

Figure 5 shows hatched dead-cell rates. Overall, 290 of 1,728 thesis cells (16.8%) are hatched after LLM. Hierarchical H continuous is 27.8% dead; polarised He continuous is 30.6% dead. Those generators contribute fewer scored events. The hatch pattern is part of the result, not a footnote.

5 Homogeneous versus heterogeneous identity

Identity alignment is the remapping of Pfeffer factor 5 (lack of diversity) into a persona mix (Pfeffer, Zorbach, and Carley, 2014). H cells use one persona on all eight nodes. He cells use a documented eight-identifier mix. Figure 6 plots collapsed live-cell means for each instrument.

Collapsed He is higher than collapsed H on both instruments (continuous +0.1584; dual-discrete +0.3986). That ordering is the opposite of a pooled “hetero buffers firestorms” claim of the sort associated with mixed CIKM chains (Maurya et al., 2025). The H mean mixes conspiracy personas (continuous about 2–2.5) with scientists (continuous near 0). The He mean mixes conspiracy-heavy mix_00/mix_01 with conspiracy-free mix_02. Composition, not the H/He label, is what moves MPR in this harvest.

Dead He cells are not evidence that a mix immunises a network. They are hatched after real

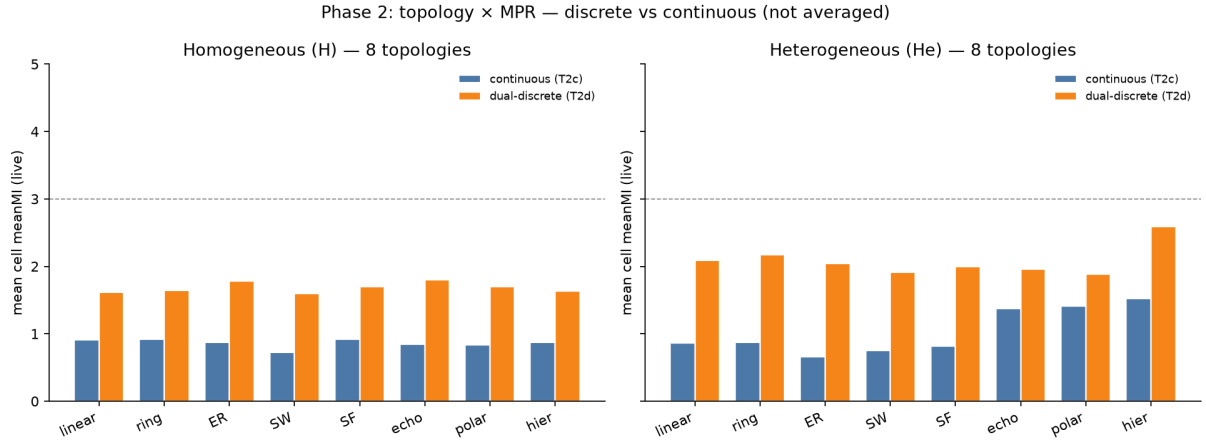


Figure 3: Live-cell mean meanMI by topology, scoring mode, and arm. Left: homogeneous. Right: heterogeneous. Discrete (T2d) and continuous (T2c) bars are not averaged. Dashed line in later figures marks the propaganda threshold $\text{MI} = 3$; this panel uses the raw 0–5 scale. $N = 1$. Source: `fig01_topology_mpr_discrete_vs_continuous.png`.

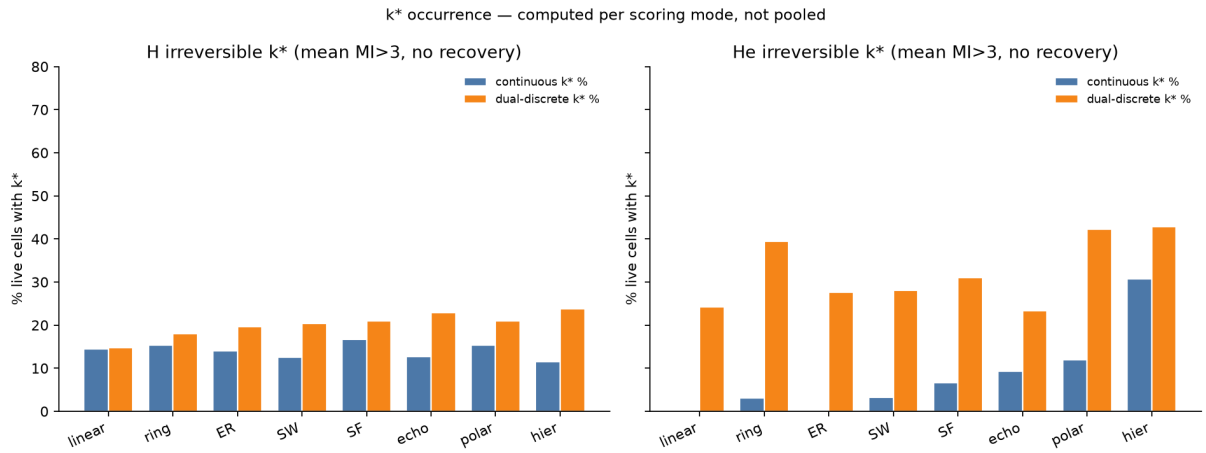


Figure 4: Share of live cells with irreversible network k^* (mean $\text{MI} > 3$ with no later recovery), by topology and scoring mode. Dual k^* uses the discrete series; continuous k^* uses the continuous series. The two rates are not added. Right-censored at eight ticks. $N = 1$. Source: `fig09_kstar_rates.png`.

LLM calls, as in Figure 5.

5.1 Homogeneous personas

Figures 7 and 8 and Table 5 report the twelve H personas collapsed across topologies. Conspiracy-family cells are the high-MPR group on both instruments (discrete 3.1691, continuous 2.2194). Science/environment is lowest on continuous (0.1108) but not uniformly lowest on dual-discrete (0.8687 versus climate-action 1.0488). Scientist and journalist personas often sit near 0 on continuous and still pick up discrete points on some articles (Figure 9). That disagreement is a mode effect. It is not evidence that “the scientist believed chemtrails”.

Figure 9 is the persona × article heatmap analogue of the previous paper’s identity-by-story grid. Figure 10 is the persona × topology grid. Conspiracy rows stay dark on both instruments. Science rows stay pale on continuous and only moderately darker on discrete. Article columns also differ: `polar_bears` is near zero on continuous H, while `climate_consensus` and the

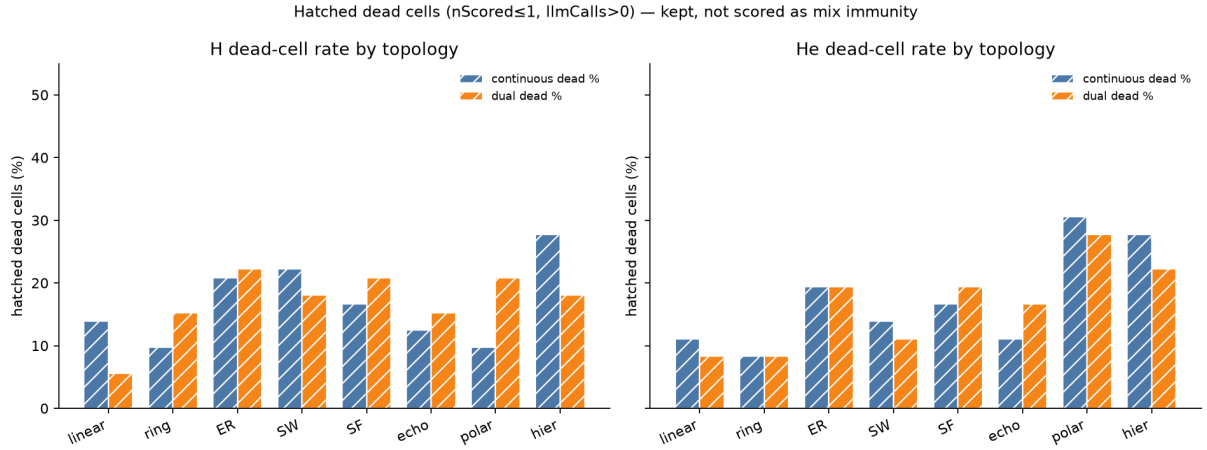


Figure 5: Hatched dead-cell rate by topology, arm, and scoring mode. Dead means $nScored \leq 1$ after real LLM calls. Hatched cells are kept and are not scored as mix immunity. $N = 1$. Source: `fig08_dead_cell_rates.png`.

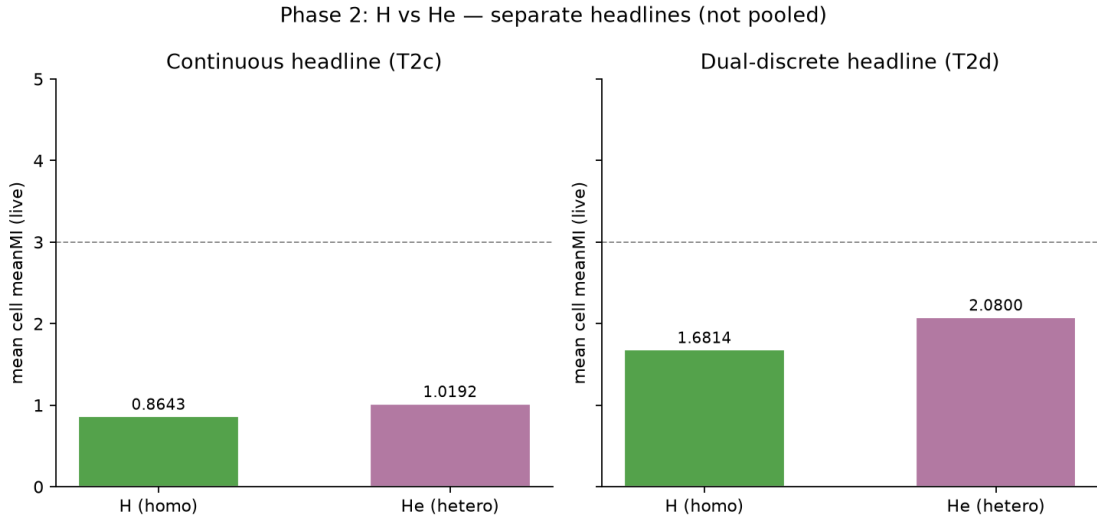


Figure 6: Collapsed live-cell meanMI for homogeneous (H) versus heterogeneous (He) arms, separately for continuous (T2c) and dual-discrete (T2d) headlines. The two panels are not averaged. $N = 1$. Source: `fig02_h_vs_he_mpr.png`.

chemtrails–Gates article sit higher. Section 6 returns to that valence contrast as Design C.

5.2 Heterogeneous mixes

Figures 11 and 12 and Table 6 report the six He mixes. `mix_00` (four conspiracy + four climate/environment) and `mix_02` (zero conspiracy + eight science/climate) are the useful contrast. Continuous: `mix_00` 1.7712 versus `mix_02` 0.2645. Dual-discrete: `mix_00` 3.1321 versus `mix_02` 1.1882. If `mix_02` is lower, that is “this conspiracy-free mix scored lower in these $N = 1$ runs”, not “diversity always stops firestorms”.

`mix_01` (two conspiracy + six climate/environment) sits between `mix_00` and `mix_02` on both instruments. `mix_03` (one conspiracy + seven other) is enough, in this harvest, to lift dual-discrete MPR to 2.0410 and continuous MPR to 0.8945. `mix_04` and `mix_05` replace core conspiracy identifiers with conspiracy-adjacent extras; they do not form a dose–response law

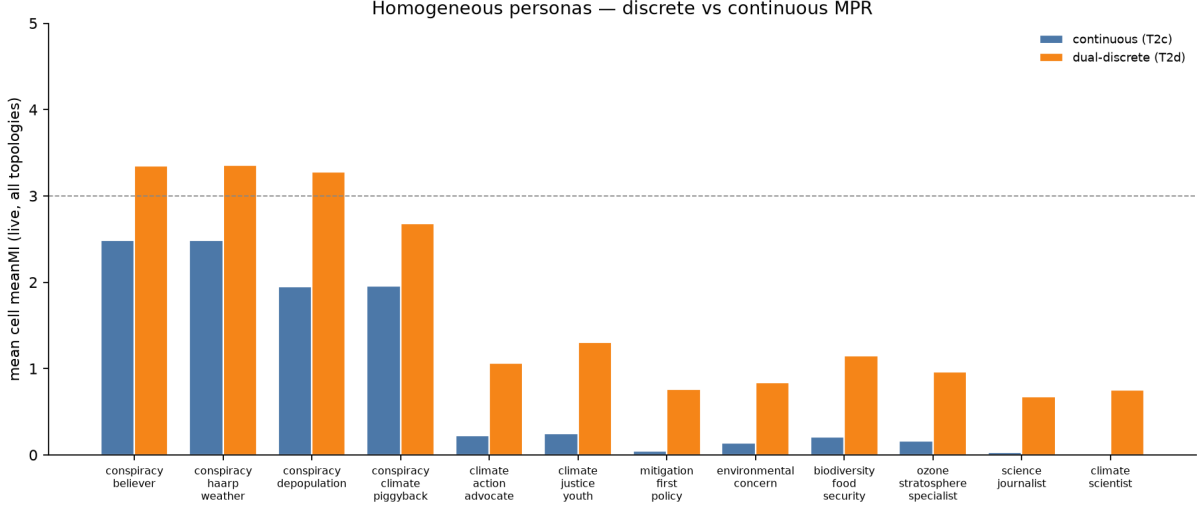


Figure 7: Homogeneous personas: live-cell mean meanMI collapsed across eight topologies, discrete versus continuous. $N = 1$, 48 cells per persona per mode before hatching. Source: `fig03_persona_mpr_bars.png`.

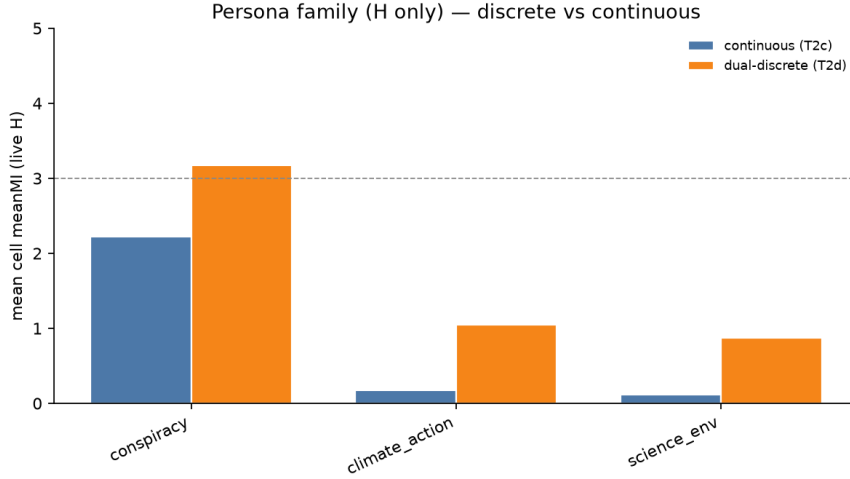


Figure 8: Persona families on homogeneous cells: conspiracy ($n = 4$), climate action ($n = 3$), science/environment ($n = 5$). Discrete and continuous remain separate. $N = 1$. Source: `fig12_persona_family.png`.

with `mix_00`.

Dual k^* on `mix_00` and `mix_01` exceeds 57%. `mix_02` has dual k^* 0.0% and continuous k^* 0.0%. The expert-inclusive mix never locked network-mean $\text{MI} > 3$ in these live cells. That is a descriptive zero on eight ticks, not a proof that experts prevent propaganda.

6 Four comparison designs

The previous paper organised empirical comparison around homogeneous versus heterogeneous identity, hop-limited trajectories, persona-conditioned heatmaps, and networked conditions (Maurya et al., 2025). The thesis protocol names four designs that this climate campaign can instantiate without reprinting crime-news cells. Phase 2 reports those four designs on the live harvest. Belief-correction injection (Phase 1 Experiment B, $n \in \{1, 3\}$) was not repeated here; Design C below is therefore seed valence, which Phase 2 did sweep.

Table 3: Homogeneous (H) topology headlines. Each topology has 72 cells per mode (12 personas $\times$ 6 articles). Live means exclude hatched dead cells.

Topology	n live dual	Disc. MPR	k^* dual	Dead dual	n live cont.	Cont. MPR	k^* cont.	Dead cont.
linear chain	68/72	1.6127	14.7%	5.6%	62/72	0.9129	14.5%	13.9%
ring	61/72	1.6402	18.0%	15.3%	65/72	0.9160	15.4%	9.7%
random ER	56/72	1.7809	19.6%	22.2%	57/72	0.8736	14.0%	20.8%
small world	59/72	1.5977	20.3%	18.1%	56/72	0.7215	12.5%	22.2%
scale-free	57/72	1.7034	21.1%	20.8%	60/72	0.9205	16.7%	16.7%
echo chamber	61/72	1.7984	23.0%	15.3%	65/72	0.8233	12.3%	9.7%
polarised	57/72	1.7005	21.1%	20.8%	65/72	0.8356	15.4%	9.7%
hierarchical	59/72	1.6315	23.7%	18.1%	52/72	0.8751	11.5%	27.8%

Table 4: Heterogeneous (He) topology headlines. Each topology has 36 cells per mode (6 mixes $\times$ 6 articles).

Topology	n live dual	Disc. MPR	k^* dual	Dead dual	n live cont.	Cont. MPR	k^* cont.	Dead cont.
linear chain	33/36	2.0852	24.2%	8.3%	32/36	0.8638	0.0%	11.1%
ring	33/36	2.1768	39.4%	8.3%	33/36	0.8760	3.0%	8.3%
random ER	29/36	2.0412	27.6%	19.4%	29/36	0.6593	0.0%	19.4%
small world	32/36	1.9096	28.1%	11.1%	31/36	0.7545	3.2%	13.9%
scale-free	29/36	1.9924	31.0%	19.4%	30/36	0.8157	6.7%	16.7%
echo chamber	30/36	1.9632	23.3%	16.7%	32/36	1.3775	9.4%	11.1%
polarised	26/36	1.8830	42.3%	27.8%	25/36	1.4137	12.0%	30.6%
hierarchical	28/36	2.5934	42.9%	22.2%	26/36	1.5242	30.8%	27.8%

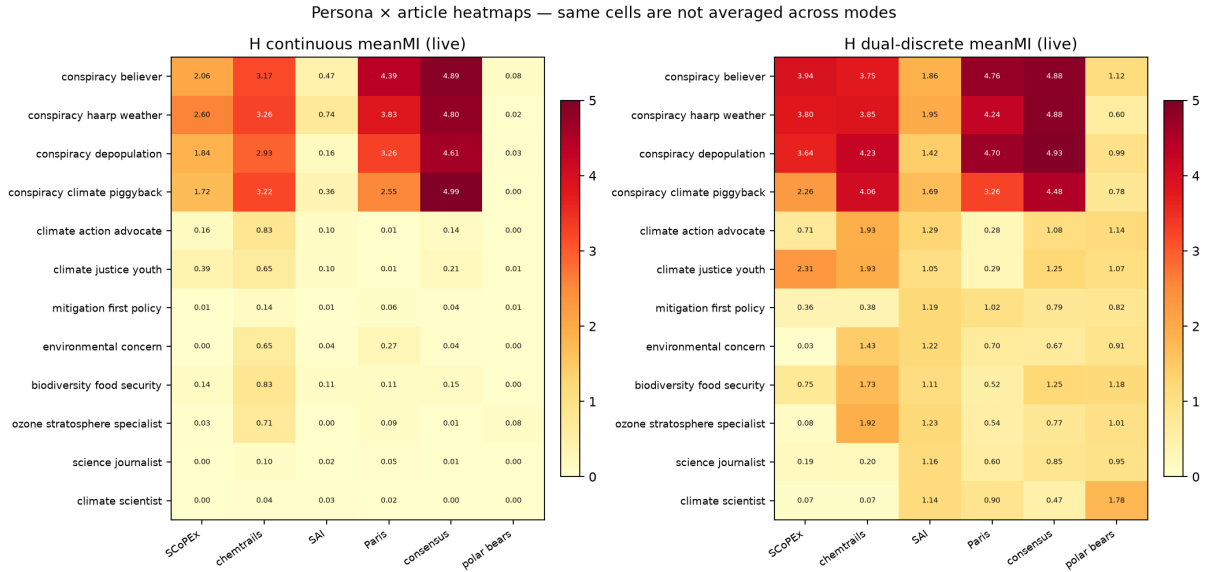


Figure 9: Homogeneous persona $\times$ article heatmaps of live-cell meanMI. Left: continuous. Right: dual-discrete. The two panels are not averaged. $N = 1$. Source: fig05_persona_article_heatmaps.png.

Table 5: Homogeneous personas collapsed across topologies. Family labels are analysis groupings, not Debnath HDBSCAN clusters.

Persona	Family	Disc. MPR	Cont. MPR	k^* dual	k^* cont.
conspiracy_believer	conspiracy	3.3522	2.4842	65.8%	47.7%
conspiracy_haarp_weather	conspiracy	3.3571	2.4829	70.5%	47.5%
conspiracy_depopulation	conspiracy	3.2823	1.9535	59.5%	34.9%
conspiracy_climate_piggyback	conspiracy	2.6838	1.9570	42.5%	31.7%
climate_action_advocate	climate action	1.0665	0.2258	0.0%	0.0%
climate_justice_youth	climate action	1.3072	0.2454	2.2%	0.0%
mitigation_first_policy	climate action	0.7571	0.0464	0.0%	0.0%
environmental_concern	science/env.	0.8359	0.1423	0.0%	0.0%
biodiversity_food_security	science/env.	1.1470	0.2041	0.0%	0.0%
ozone_stratosphere_specialist	science/env.	0.9639	0.1672	0.0%	0.0%
science_journalist	science/env.	0.6745	0.0310	0.0%	0.0%
climate_scientist	science/env.	0.7517	0.0120	0.0%	0.0%

Table 6: Heterogeneous mixes collapsed across topologies.

Mix	Composition	Disc. MPR	Cont. MPR	k^* dual	k^* cont.
mix_00	4 conspiracy + 4 climate/env.	3.1321	1.7712	57.5%	12.8%
mix_01	2 conspiracy + 6 climate/env.	2.7795	1.4770	58.5%	9.8%
mix_02	0 conspiracy + 8 science/climate	1.1882	0.2645	0.0%	0.0%
mix_03	1 conspiracy + 7 other	2.0410	0.8945	33.3%	10.3%
mix_04	2 conspiracy-adj. + 6 other	1.6896	0.7021	20.5%	5.3%
mix_05	2 conspiracy-adj. + 6 other	1.6713	1.0134	23.3%	7.5%

Each design is a contrast, not a significance test. $N = 1$ supplies point estimates only.

6.1 Design A: linear-chain analogue

Design A repeats the CIKM topology idea — a linear chain — on the climate corpus (Maurya et al., 2025). Homogeneous chains assign one persona; heterogeneous chains cycle a mix. The engine object is `linear_chain` with eight nodes and drip seed `node_0`. The hop budget is eight, not CIKM 30.

Tables 3 and 4 already contain the chain rows. H continuous chain MPR is 0.9129 ($n = 62$ live of 72). H dual-discrete chain MPR is 1.6127 ($n = 68$). He continuous chain MPR is 0.8638 ($n = 32$ of 36). He dual-discrete chain MPR is 2.0852 ($n = 33$).

On the continuous instrument, the He chain sits slightly below the H chain (0.8638 versus 0.9129). On the discrete instrument, the He chain sits above the H chain (2.0852 versus 1.6127). The two instruments disagree on the CIKM-style ordering for this topology. A climate-domain claim that “heterogeneous chains buffer” therefore fails already at the instrument split, before any article-wise majority test. `mix_02` inside the He collapse is the expert-inclusive mix that the CIKM analogue requires; the He chain mean still includes `mix_00`. Composition again dominates the label.

Design A is eligible as a protocol (climate chains exist). It is not a successful conceptual replication of the CIKM homo>hetero severity pattern under the criteria in the discovery notes: six articles rather than ≥ 10 , $N = 1$, `gpt-4o-mini` only, and mode disagreement on the collapsed chain.

Table 7: Article (valence) collapse. Articles are a factor, not statistical replicates. $N = 1$.

Article	H discrete	H continuous	He discrete	He continuous
scopex_2017	1.5390	0.7321	2.3718	0.7694
chemtrails_gates_2018_2021	2.1383	1.3585	2.3885	1.7852
sai_geoengineering	1.3576	0.1946	1.1160	0.1523
paris_agreement	1.7321	1.3334	2.4591	1.6821
climate_consensus	2.2610	1.5721	2.8991	1.8896
polar_bears	1.0399	0.0166	1.1444	0.0415

6.2 Design B: networked topology and k^*

Design B is the networked expansion: eight generators $\times$ identity $\times$ two instruments, with irreversible k^* as the firestorm clock. Section 4 is the full table. The design question is whether hubs, chambers, or hierarchy move live MPR and k^* relative to a short path or an ER null, holding the seed set fixed.

No generator dominates both instruments and both arms. Scale-free is highest on H continuous and mid-pack on He continuous. Echo chamber is highest on H dual-discrete and mid-pack on He dual-discrete. Hierarchical He is highest on both He instruments and also among the deadliest cells. ER He is lowest on continuous MPR (0.6593) with continuous k^* 0.0% on 29 live cells, yet dual-discrete ER He remains 2.0412 with k^* 27.6%.

Design B therefore supports only a weak descriptive statement. Topology rank is mode-dependent and arm-dependent. Hub generators do not uniquely produce textbook k^* in this $N = 1$ eight-tick window. The statement “scale-free causes firestorms” is not licensed.

6.3 Design C: seed valence

Design C holds topology and mix in the collapse and contrasts articles. Pfeffer, Zorbach, and Carley (2014) describe firestorm messages as affectively charged; they do not number a factor called valence. The campaign treats article wording as the valence knob: calm SCoPEX measurement text versus high-arousal chemtrails–Gates text, plus four further climate items.

Table 7 reports live-cell means by article, arm, and instrument. On both arms, `climate_consensus` and `chemtrails_gates_2018_2021` sit at the high end. `polar_bears` sits at the low end, especially on continuous H (0.0166) and He (0.0415). `sai_geoengineering` is low on continuous and only moderate on discrete. `scopex_2017` is not uniformly the calmest cell: H discrete 1.5390 exceeds H discrete polar bears (1.0399), while H continuous SCoPEX (0.7321) far exceeds H continuous polar bears.

The predicted H1 direction (`chemtrails` > SCoPEX on the same cells) holds in this collapse for both arms and both instruments. Chemtrails H continuous 1.3585 exceeds SCoPEX H continuous 0.7321; chemtrails H discrete 2.1383 exceeds SCoPEX H discrete 1.5390. The same sign appears on He. The contrast still confounds topic, frame, and prior conspiracy fit. It is compatible with a valence reading. It is not a test that isolated emotional charge.

6.4 Design D: closed-group wiring

Design D asks whether echo-chamber and polarised generators concentrate distortion relative to open graphs (ER, small-world, chain) on the same identity mix. Pfeffer factor 3 (network clusters) and parts of factor 4 (unrestrained flow) and factor 5 (homophily) are the 2014 sources for this remapping (Pfeffer, Zorbach, and Carley, 2014). Phase 1’s extra echo cell died at tick 1 with zero LLM calls. Phase 2 completed echo-chamber and polarised cells on both arms; those cells can be described. They still cannot be sold as emergent echo, because the generators *input* chambers.

On homogeneous continuous cells, echo (0.8233) and polarised (0.8356) sit below scale-free (0.9205) and even below the linear chain (0.9129). Closed wiring does not raise H continuous MPR in this harvest. On homogeneous dual-discrete cells, echo is the highest generator (1.7984). The instrument split again forbids a single echo claim.

On heterogeneous continuous cells, echo (1.3775), polarised (1.4137), and hierarchical (1.5242) sit well above ER (0.6593) and the linear chain (0.8638). Closed or tiered wiring coincides with higher He continuous MPR here. Polarised He is also 30.6% dead on continuous, so the live mean is a selected subset. He dual-discrete does not repeat the same ranking: hierarchical leads (2.5934), and polarised is lowest (1.8830) among the eight.

Design D therefore yields a mode- and arm-conditional description. He continuous cells on echo and polarised graphs scored higher than He continuous cells on ER in this $N = 1$ draw. H continuous cells did not. The Phase 1 dead echo cell is not recycled as a negative finding; the Phase 2 echo cells are live enough to describe and still too weakly replicated to confirm H5.

7 Pooled-mean check

A single campaign MPR is a tempting summary. Three pooled scalars are easy to compute from Table 2. All three are invalid as headlines. This section prints them so that they are not rediscovered later as “the Phase 2 result”.

Invalid pool 1: discrete with continuous. The arithmetic mean of H continuous 0.8608 and H dual-discrete 1.6814 is 1.2711. That number has no auditor. It mixes a float instrument with a discrete instrument. Dual-discrete already sits higher on every topology (Figure 3). Averaging the two columns erases the scoring-mode effect that Phase 2 was built to expose.

Invalid pool 2: H with He on one instrument. Weighting live cells, the H+He continuous mean is

$$(482 \times 0.8608 + 238 \times 1.0192)/720 = 0.9132.$$

The H+He dual-discrete mean is

$$(478 \times 1.6814 + 240 \times 2.0800)/718 = 1.8146.$$

Each scalar hides the composition problem in Section 5. Conspiracy-heavy mixes pull He up; scientist personas pull H down. A reader who sees only 0.9132 cannot recover that `mix_02` continuous MPR is 0.2645 and conspiracy-believer continuous MPR is 2.4842.

Invalid pool 3: “hetero buffers” from the wrong collapse. CIKM-style buffering is a claim about mixed identity on a matched topology and article, not about averaging twelve H personas with six He mixes (Maurya et al., 2025). Collapsed He > collapsed H on both instruments in Table 2. A pooled story that “diversity helped” would have to ignore that sign, or restrict the contrast to `mix_02` versus conspiracy H, which is Design A/C identity rather than a grand mean.

What the check allows. Report four headlines: H continuous, H dual-discrete, He continuous, He dual-discrete. Report family and mix collapses beside them. Report topology ranks inside each of those four. Do not promote 1.2711, 0.9132, or 1.8146 to the abstract.

8 D-net, Pfeffer observables, and Debnath

The D-net arm imports a reconstructed graph and runs the same two scoring modes on homogeneous conspiracy and mixed identities. The empirical file is `thesisExperiment/data/derived/debnath_hashtag_cascade.json`. Kind: hashtag co-occurrence. `notARetweetCascade: true`. Nodes are users keyed by published hashtags (63 nodes, 228 directed edges). Edges reuse a `retweets[]` container only as a directed-edge slot. The graph is not a hydrated retweet cascade and not a FakeNewsNet conversation tree.

Hydration of Debnath’s tweet identifiers failed. Mendeley `10.17632/546hsym93p.1` is an identifier dump, not tweet text (Debnath et al., 2023a). OSF `dataset.csv` stores `tweet_id` in Excel scientific notation, so low digits are gone; usable digit identifiers in the sampled prefix were zero. No Twitter/X bearer was present. 814,924 tweets were not hydrated. Belief profiles remain theory-faithful reductions, not skip-gram/HDBSCAN centroids (Debnath et al., 2023b; Debnath, 2023).

Debnath reports no empirical 0–5 MPR. Simulated auditor MI is therefore not Twitter MI. Ruths and Pfeffer (2014) already warn that platform traces are not a transparent sample of behaviour; this comparison inherits that caution twice, because the empirical object is a hashtag fallback.

8.1 Structural comparison

Figure 13 and Table 8 compare empirical hashtag-graph metrics with the mean of 16 simulated D-net cascades. Structural similarity (share of compared metrics within a 30% band) is 0.6885: breadth matches, depth and structural virality (Goel et al., 2016) do not. Size is informational only.

Kolmogorov–Smirnov tests use $n_{\text{real}} = 1$ and $n_{\text{sim}} = 16$. Those tests are underpowered. KS p -values (depth 0.1104, breadth 0.2954, virality 0.1104) are not a twin-validation claim. Jensen–Shannon divergence equals 1 on all three metrics. DTFS = 0.2754 with `isValidated=false` (threshold 0.70). Content correlation is set to 0 because Debnath has no empirical MPR.

Table 8: D-net structural metrics. Match = within a 30% band of the empirical value. KS/JS: $n_{\text{real}} = 1$, underpowered.

Metric	Emp.	Sim. mean	Match (30%)	Note
depth	4	2.5625	no	longest path from root
breadth	33	40.9375	yes	max nodes at one level
size	61	53.5	—	informational
struct. virality	2.7	1.7963	no	Goel et al. (2016)
speed	n/a emp.	7 ticks	—	do not convert ticks to hours
mean MI	no emp. MPR	2.5529	do not equate	auditor only

Table 9: D-net auditor headlines on the hashtag co-occurrence graph. Sidecar continuous on dual runs is not the T2c headline.

Run	Article	Mode	Disc. MPR	Cont. MPR
Dnet_c_H_conspiracy	scopex_2017	continuous	—	1.7349
Dnet_c_H_conspiracy	chemtrails–Gates	continuous	—	3.3378
Dnet_c_He_mixed	scopex_2017	continuous	—	0.9983
Dnet_c_He_mixed	chemtrails–Gates	continuous	—	2.5068
Dnet_d_H_conspiracy	scopex_2017	dual-disc.	4.2537	sidecar 2.2495
Dnet_d_H_conspiracy	chemtrails–Gates	dual-disc.	3.7303	sidecar 3.1251
Dnet_d_He_mixed	scopex_2017	dual-disc.	2.6880	sidecar 1.1302
Dnet_d_He_mixed	chemtrails–Gates	dual-disc.	1.5297	sidecar 1.1497

8.2 Simulated D-net auditor scores

Table 9 lists LLM-as-judge headlines on the custom graph. They are not Twitter MPR. Continuous D-net conspiracy $\times$ chemtrails is 3.3378; the same graph $\times$ SCoPEX is 1.7349. Dual-discrete conspiracy $\times$ SCoPEX is 4.2537 with sidecar continuous 2.2495. Mixing those columns would repeat Invalid pool 1 on a 63-node graph.

8.3 Pfeffer 2014: seven factors, then a remapping

Pfeffer, Zorbach, and Carley (2014) list seven interrelated factors in the Outlook section:

1. speed and volume of communication;
2. binary choices (like/share/retweet as either–or);
3. network clusters;
4. unrestrained information flow;
5. lack of diversity;
6. cross-media dynamics;
7. network-triggered decision processes.

That list is the primary theory. The thesis six-knob vocabulary (valence, surprise, identity, clustering, echo, temporal acceleration) is a computational remapping, not a restatement of the 2014 paper. Table 10 records seven *observables* so that the operational table matches the paper’s count. Cross-media remains held: no legacy-media broadcast agent exists in the engine. Binary choices (factor 2) have no dedicated row, because the engine is ternary (forward/reinterpret/drop).

Simulated identity homophily of 1 on D-net is partly tautological for the homogeneous conspiracy run. It does not demonstrate that Pfeffer factor 5 was tested and confirmed. Surprise and temporal acceleration were held. The table is an honesty ledger, not a seven-factor confirmation.

Table 10: Pfeffer observables on the Debnath hashtag fallback versus simulated D-net. Empirical valence is a toxicity/hashtag proxy, not MPR. Simulated mean MI is an auditor score, not Twitter MI.

Observable	Pfeffer 2014 map	Empirical	Simulated
valence	affective character of firestorm messages (definition), not a numbered Outlook name	0.1097 (paper mean toxicity 0.17)	mean MI 2.5529 (auditor)
surprise	#1 speed/volume (shock vs drip)	paper notes SCoPEX spike; not reconstructed	held (drip)
identity	#5 lack of diversity	conspiracy share 0.4444 / other 0.5556	conspiracy share 0.7222
clustering	#3 network clusters	$n = 63$, mean degree 7.2381, $Q = 0.4039$	$n = 63$, mean degree 7.2381, $Q = 0.4566$
echo	#3 + #4 + parts of #5	identity homophily 0.864	edge homophily 1; identity homophily 1
temporal	#1 / #7	not available on a hashtag fallback	7 ticks (sim clock)
cross-media	#6	held	held (no engine knob)

9 Limitations

The harvest itself marks this campaign as not thesis-grade (**thesisGrade: false**). The following limits are part of the result, not an appendix apology.

$N = 1$. One graph seed (42), one run per config. No error bars, no topology significance tests, no permutation tests. Events on the same node share inbox and trust state; nodes share a graph. Treating scored events as i.i.d. replicates would inflate precision.

Eight hops are not CIKM 30. The hop/tick budget is a logged cost compression versus the CIKM chain length of 30 (Maurya et al., 2025). It is not Debnath’s hop protocol and not an empirical firestorm duration. k^* irreversibility is right-censored at eight ticks.

Debnath’s eight is a skip-gram window. Debnath et al. (2023b) define a context window of eight *words* for skip-gram embeddings. That hyperparameter is unrelated to this campaign’s eight hops.

Hatched dead cells: 290. Dead cells are kept. They are not zeros and they are not mix immunity. Live-only means drop those cells and therefore describe surviving cascades. Hierarchical and some polarised He slices are particularly incomplete.

Two instruments. Dual-discrete headline $\neq$ continuous headline. Dual gap $\neq$ MPR. Section 7 lists the pooled scalars that must not re-enter the abstract.

Hashtag graph, not a retweet cascade. The Debnath reconstruct is a hashtag co-occurrence fallback. 814k tweet identifiers were not hydrated (Debnath et al., 2023a). BPs are theory-faithful reductions, not HDBSCAN centroids. No empirical MPR exists on Debnath, so simulated MI cannot “match Twitter misinformation”.

KS/JS with $n_{\text{real}} = 1$. Distributional tests cannot support a digital-twin claim. DTFS failed the 0.70 threshold (0.2754).

Pfeffer 2014 has seven factors. The thesis six-knob list is a remapping (Pfeffer, Zorbach, and Carley, 2014). Cross-media is held (no engine knob). Surprise is held as drip. Binary like/share is not the engine’s ternary action set.

Role-play is not a field community. gpt-4o-mini system prompts can produce prompt-faithful conspiracy language (Park et al., 2023). High conspiracy MPR is compatible with obedience to those prompts. It is not, by itself, evidence of stubborn belief or of chemtrails communities in the wild. False news online does travel farther than true news in observational cascades (Vosoughi, Roy, and Aral, 2018); this simulator does not re-estimate that fact.

Auditor as truth. Five author-written QA items define MI. A different key would move the numbers. Ground truth is the curated scientific record in the article pack, not a moral ranking of personas. Human-eval templates exist in run directories and were not completed as an IRB study.

Phase 1 is a different campaign. The discrete 12×12 chain tables under `thesisExperiment/results/tables/` are not spliced into these headlines. Fact-check injection (Experiment B) and the distortion keyword heuristic (Experiment C) belong to that earlier drop-in set unless a later pass re-runs them under `runs_phase2/`.

Correction literature is out of scope here. Continued-influence and debiasing results (Lewandowsky et al., 2012) motivated Phase 1 Experiment B. Phase 2 does not estimate recovery.

10 Chapter conclusion

Phase 2 completed the eight-topology $\times$ H/He $\times$ dual/continuous grid (1,728 cells, 0 missing) plus four D-net configs. The harvest is a real-gpt-4o-mini $N = 1$ pilot with `thesisGrade=false`.

Four headlines stand, and they are not interchangeable. Homogeneous continuous MPR is 0.8608 ($n = 482$ live). Homogeneous dual-discrete MPR is 1.6814 ($n = 478$). Heterogeneous continuous MPR is 1.0192 ($n = 238$). Heterogeneous dual-discrete MPR is 2.0800 ($n = 240$). Dual-discrete sits higher than continuous on every topology. That is a scoring-mode effect.

Collapsed He exceeds collapsed H on both instruments. Composition explains the sign: conspiracy personas and conspiracy-heavy mixes score high; science personas and `mix_02` score low. The H/He label is not a buffering law.

Topology rank depends on instrument and arm. No generator is “the firestorm topology” in this draw. Echo and polarised He continuous cells scored higher than ER He continuous cells; H continuous cells did not repeat that pattern. Hierarchical He leads both He instruments and is heavily hatched.

Design A (climate chains) is protocol-complete and inconclusive as a CIKM conceptual replication: the two instruments disagree on the collapsed chain ordering, and six articles with $N = 1$ do not meet the pre-registered success bar. Design B (networked k^*) shows mode-dependent ranks and weak irreversibility on an eight-tick clock. Design C (valence) shows chemtrails $>$ SCoPEX in the article collapse, still confounded with topic and prior fit. Design D

(closed-group wiring) is describable because Phase 2 echo cells lived; it does not confirm emergent echo.

The D-net comparison is structural. Breadth lies within a 30% band of the hashtag graph; depth and structural virality do not. KS tests with one empirical graph are underpowered. DTFS fails. Simulated MI is not Twitter MPR. Pfeffer, Zorbach and Carley list seven Outlook factors; this campaign remaps six knobs and holds cross-media.

What an examiner can accept is therefore narrow. The engine produces non-zero, persona-sensitive auditor scores on climate seeds. Conspiracy-conditioned nodes sit far above scientist-conditioned nodes on both instruments in this $N = 1$ harvest. Discrete and continuous campaigns must be reported as two leagues. Everything else — topology winners, hetero buffering, Twitter matching, seven-factor confirmation — remains outside what these cells can carry.

References

- Barabási, A.-L. and R. Albert (1999). “Emergence of Scaling in Random Networks”. In: *Science* 286.5439, pp. 509–512. DOI: 10.1126/science.286.5439.509.
- Debnath, R. et al. (2023a). *Conspiracy Spillovers and Geoengineering — Tweet ID Dataset*. Mendeley Data. Tweet IDs; hydration requires a platform API. DOI: 10.17632/546hsym93p.1.
- Debnath, R. (2023). *geoeng: Analysis Codes for Conspiracy Spillovers and Geoengineering*. Hashtag extraction, skip-gram embeddings, NRC, Perspective; not the tweet dump.
- Debnath, R., D. M. Reiner, B. K. Sovacool, F. Müller-Hansen, T. Repke, R. M. Alvarez, and S. D. Fitzgerald (2023b). “Conspiracy Spillovers and Geoengineering”. In: *iScience* 26.3. PMC10040962, p. 106166. DOI: 10.1016/j.isci.2023.106166.
- Erdős, P. and A. Rényi (1960). “On the Evolution of Random Graphs”. In: *Publications of the Mathematical Institute of the Hungarian Academy of Sciences* 5, pp. 17–61.
- Goel, S., A. Anderson, J. Hofman, and D. J. Watts (2016). “The Structural Virality of Online Diffusion”. In: *Management Science* 62.1, pp. 180–196. DOI: 10.1287/mnsc.2015.2158.
- Kempe, D., J. Kleinberg, and É. Tardos (2003). “Maximizing the Spread of Influence through a Social Network”. In: *Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, pp. 137–146. DOI: 10.1145/956750.956769.
- Lewandowsky, S., U. K. H. Ecker, C. M. Seifert, N. Schwarz, and J. Cook (2012). “Misinformation and Its Correction: Continued Influence and Successful Debiasing”. In: *Psychological Science in the Public Interest* 13.3, pp. 106–131. DOI: 10.1177/1529100612451018.
- Maurya, R. G., V. Shukla, R. A. Dandekar, R. Dandekar, and S. Panat (2025). “Simulating Misinformation Propagation in Social Networks using Large Language Models”. In: *1st Workshop on LLM Agents for Social Simulation (LASS) at ACM CIKM 2025*. Outstanding Paper Award; ACM proceedings DOI not verified in the discovery pass. Seoul, Republic of Korea. DOI: 10.48550/arXiv.2511.10384. arXiv: 2511.10384 [cs.SI].
- Park, J. S., J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein (2023). “Generative Agents: Interactive Simulacra of Human Behavior”. In: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST ’23)*. ACM. DOI: 10.1145/3586183.3606763.

- Pfeffer, J., T. Zorbach, and K. M. Carley (2014). “Understanding Online Firestorms: Negative Word-of-Mouth Dynamics in Social Media Networks”. In: *Journal of Marketing Communications* 20.1-2, pp. 117–128. DOI: 10.1080/13527266.2013.797778.
- Ruths, D. and J. Pfeffer (2014). “Social Media for Large Studies of Behavior”. In: *Science* 346.6213, pp. 1063–1064. DOI: 10.1126/science.1257756.
- Vosoughi, S., D. Roy, and S. Aral (2018). “The Spread of True and False News Online”. In: *Science* 359.6380, pp. 1146–1151. DOI: 10.1126/science.aap9559.
- Watts, D. J. and S. H. Strogatz (1998). “Collective Dynamics of ‘Small-World’ Networks”. In: *Nature* 393, pp. 440–442. DOI: 10.1038/30918.
- Zheng, L., W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica (2023). “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”. In: *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. DOI: 10.48550/arXiv.2306.05685. arXiv: 2306.05685.

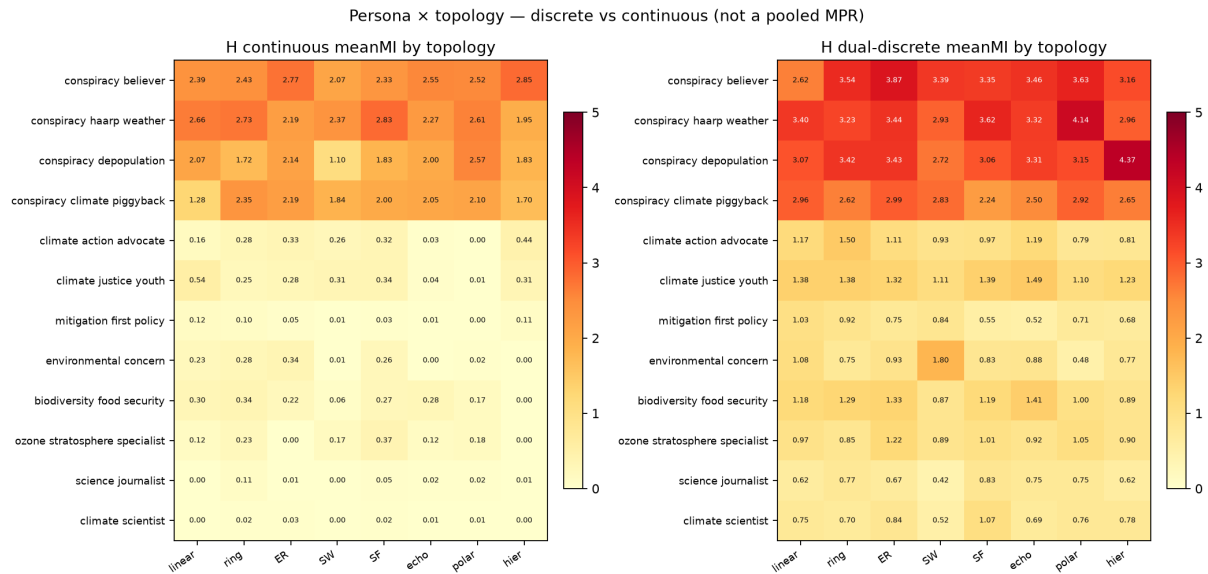


Figure 10: Homogeneous persona × topology heatmaps of live-cell meanMI. Left: continuous. Right: dual-discrete. $N = 1$. Source: fig06_persona_topology_heatmaps.png.

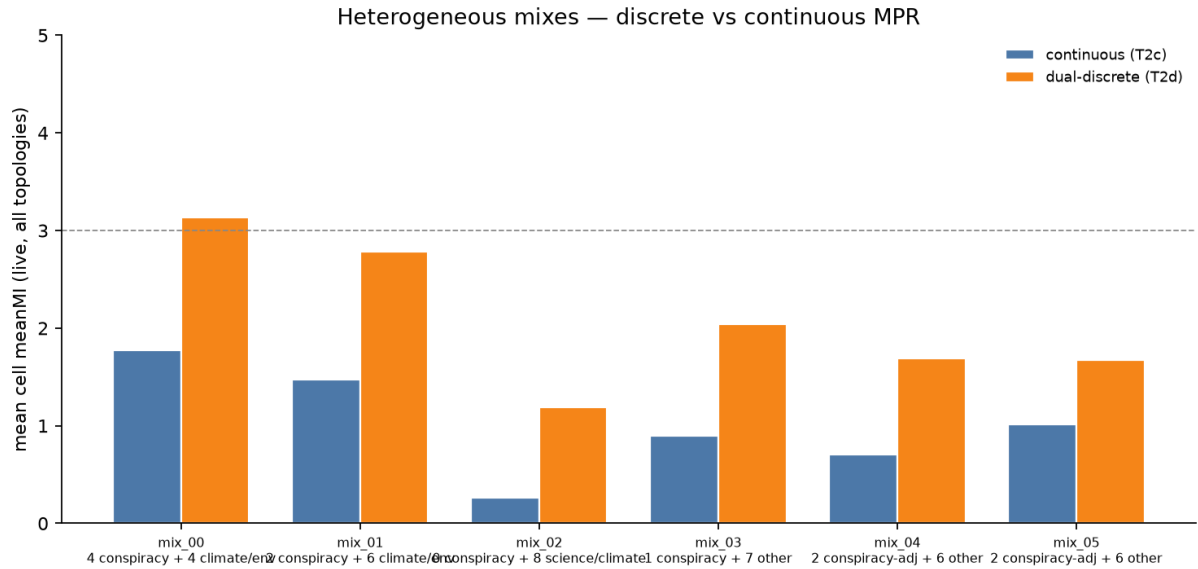


Figure 11: Heterogeneous mixes: live-cell mean meanMI collapsed across eight topologies, discrete versus continuous. $N = 1$, 48 cells per mix per mode before hatching. Source: fig04_mix_mpr_bars.png.

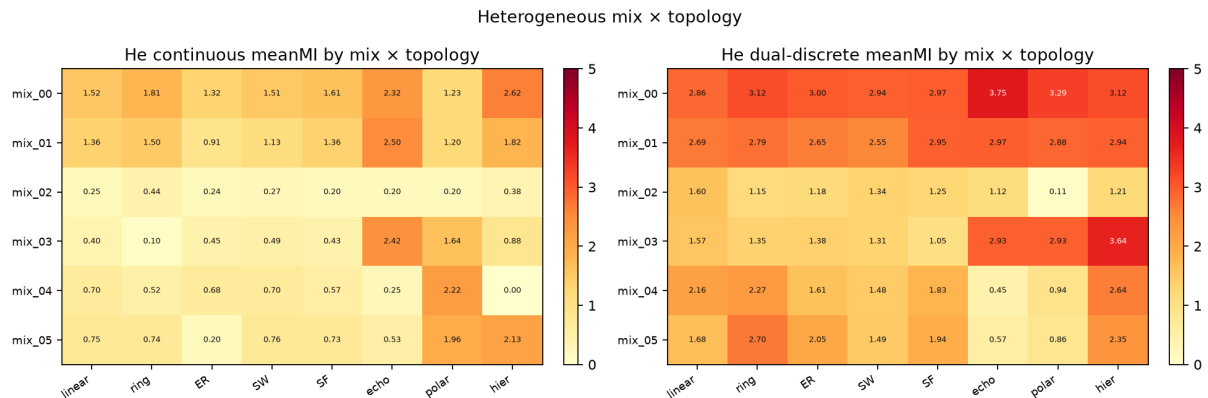


Figure 12: Heterogeneous mix × topology heatmaps of live-cell meanMI. Left: continuous. Right: dual-discrete. $N = 1$. Source: fig07_mix_topology_heatmaps.png.

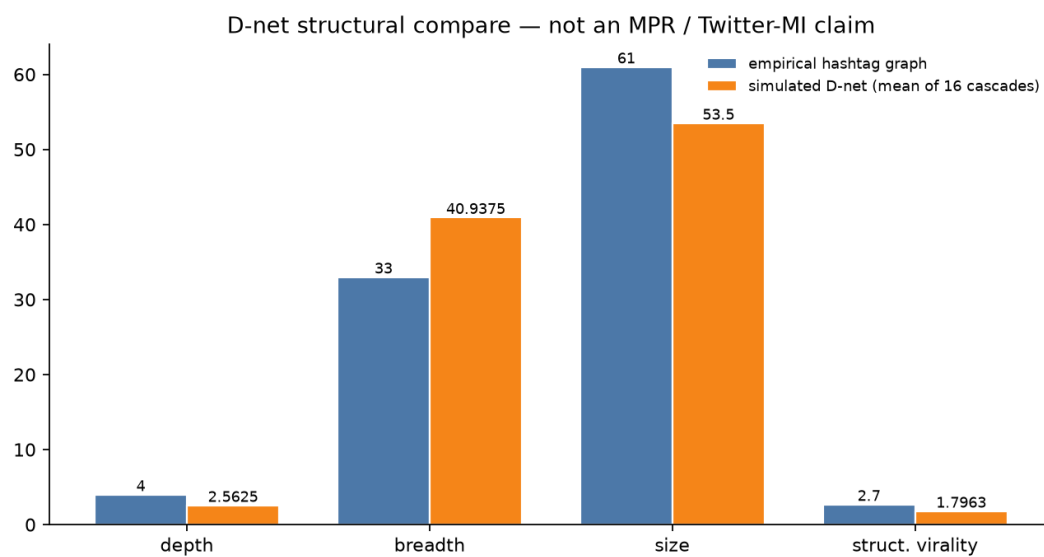


Figure 13: Empirical hashtag-graph metrics versus mean simulated D-net metrics (16 cascades). This is a structural comparison, not an MPR or Twitter-MI claim. Source: `fig11_dnet_structural.png`.